DATA SCIENCE REPORT

MOBILITY PRICING INTELLIGENCE: COST PREDICTION & PRICING DRIVER ANALYSIS

Pricing Drivers, Ride Economics & Mobility Operations



TOOLS USED

Python, Pandas, Numpy, Matplotlib, Plotly & ML libraries

PRESENTED BY

Tejas Jadhav

GITHUB

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Overview & Business Objective

1.1 PROJECT OVERVIEW

Ride-sharing platforms operate in an environment where ride costs can vary based on several factors, including expected ride duration, vehicle type, rider demand, driver availability, booking period, location, and customer characteristics.

Understanding these relationships is important for estimating ride costs consistently and identifying the operational factors associated with pricing variation.

The Ride Pricing Analytics project applies machine learning techniques to historical ride-sharing data to develop a ride-cost prediction system and identify the factors that contribute most strongly to model predictions.

The project combines:

- Data understanding and validation
- Exploratory Data Analysis
- Feature engineering
- Data preprocessing
- Regression modeling
- Hyperparameter tuning
- Ensemble learning
- Model evaluation and comparison
- Prediction-error analysis
- Model explainability
- Business-oriented interpretation

The objective is not only to predict historical ride costs, but also to understand why the models make their predictions and where significant prediction errors remain

1.2 PROBLEM STATEMENT

Ride costs can vary across different combinations of demand, supply, ride duration, vehicle type, booking period, location, and customer characteristics.

The company currently relies primarily on ride duration when determining ride fares. However, the historical dataset contains additional information that may provide useful signals for estimating ride costs.

The key problem addressed in this project is:

How can historical ride-sharing data be used to build a machine learning model that accurately predicts historical ride costs while identifying the major factors contributing to those predictions?

The project therefore treats Historical_Cost_of_Ride as the target variable and uses ride-level, customer, vehicle, booking, and demand-supply information as predictive features.

(Note: It is important to note that the dataset contains historical observed ride costs rather than an explicitly defined optimal or surge-pricing target)

1.3 PROJECT OBJECTIVES

The major objectives of the project are:

- Develop a machine learning-based ride-cost prediction system.
- Understand the underlying structure and characteristics of the data.
- Identify important pricing patterns through EDA.
- Analyze the relationship between ride duration & ride cost.
- Analyze demand and supply relationships using rider and driver information.
- Train and compare multiple regression models.
- Improve selected models using hyperparameter tuning.
- Evaluate Voting and Stacking ensemble approaches.
- Evaluate model performance using appropriate regression metrics.
- Investigate residuals and large prediction errors.
- Identify the most influential features driving model predictions.



1.4 PROJECT SCOPE

The project follows a structured workflow covering data analysis, price modeling, model explainability, and business interpretation to understand demand patterns, evaluate predictive performance, identify key demand drivers, and derive actionable retail insights.

01

Data Analysis

- Dataset inspection
- Data quality assessment
- Distribution analysis
- Relationship analysis

02

Demand Modeling

- Feature preparation
- Regression modeling
- Model comparison
- Model evaluation

03

Explainability

- Lasso coefficient analysis
- SHAP analysis
- Permutation feature importance

04

Business Interpretation

- Business Interpretation
- Inventory implications
- Pricing and promotion observations
- Seasonal and weather-related effects
- Recommendations for retail planning

Data Understanding & Exploratory Data Analysis

2.1 DATA OVERVIEW

The dataset contains 1,000 observations and includes information related to riders, drivers, location, customer loyalty, previous rides, ratings, booking time, vehicle type, expected ride duration, and historical ride cost.

Data Source: Kaggle

Dataset Size

1,000
Records

10
Original Features

Data Diversity

3
Location Categories

2
Vehicle Types

4
Time Segments

DATA OVERVIEW

Below is a detailed description of the feature set:

Dataset Features	Type	Feature Description
Number_of_Riders	Numerical (Discrete)	Number of riders associated with the observation
Number_of_Drivers	Numerical (Discrete)	Number of available drivers associated with the observation
Location_Category	Categorical	Geographic classification of the ride location
Customer_Loyalty_Status	Categorical	Loyalty category of the customer
Number_of_Past_Rides	Numerical (Discrete)	Number of previous rides completed by the customer
Average_Ratings	Numerical (Discrete)	Average rating associated with the customer
Time_of_Booking	Categorical	Time period during which the ride was booked
Vehicle_Type	Categorical	Category of vehicle selected for the ride
Expected_Ride_Duration	Numerical (Discrete)	Expected duration of the ride
Historical_Cost_of_Ride	Numerical (Continuous)	Historical cost of the ride and target variable used for regression modeling

2.2 TARGET VARIABLE

The target variable selected for prediction is:

Historical_Cost_of_Ride

Historical_Cost_of_Ride represents the observed historical cost associated with each ride.

The target has:

372.50
Mean

362.02
Median

25.99
Minimum

836.12
Maximum

187.16
Standard Deviation

The target demonstrates substantial variation across observations, making it suitable for a continuous regression problem.

2.3 DATA QUALITY & PREPARATION OVERVIEW

The initial data-quality analysis identified the following characteristics:

Missing Values

No missing values were identified across the dataset.

Duplicate Records

No duplicate records were identified.

Numerical Values

All numerical variables contained valid non-negative observations.

Categorical Variables

The categorical variables contained well-defined categories without unexpected values or extremely rare categories requiring consolidation.

Outlier Analysis

An IQR-based analysis identified 10 potential outliers in Number_of_Drivers, representing 1.0% of the dataset.

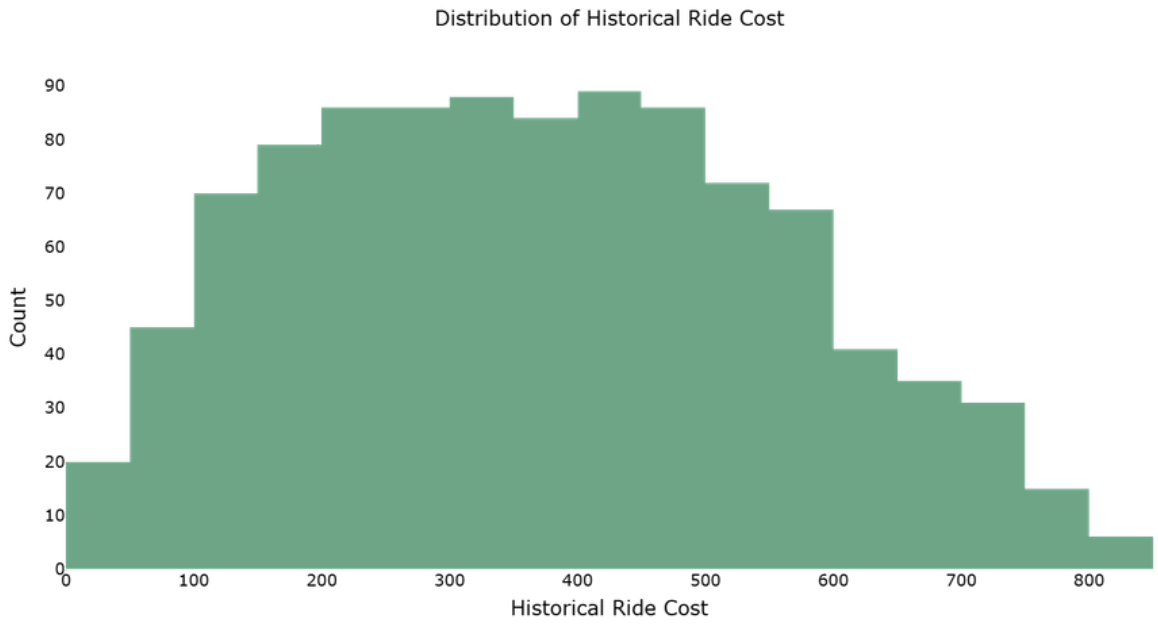
Investigation showed that these observations contained driver counts between 80 & 89 and were accompanied by valid values across the remaining variables.

Therefore, these observations were considered legitimate high-supply observations rather than data errors and were retained for modeling.

2.4 Exploratory Data Analysis

Historical ride costs are broadly distributed across a wide range, with the highest concentration occurring in the mid-range.

$372.50
Average Ride Cost



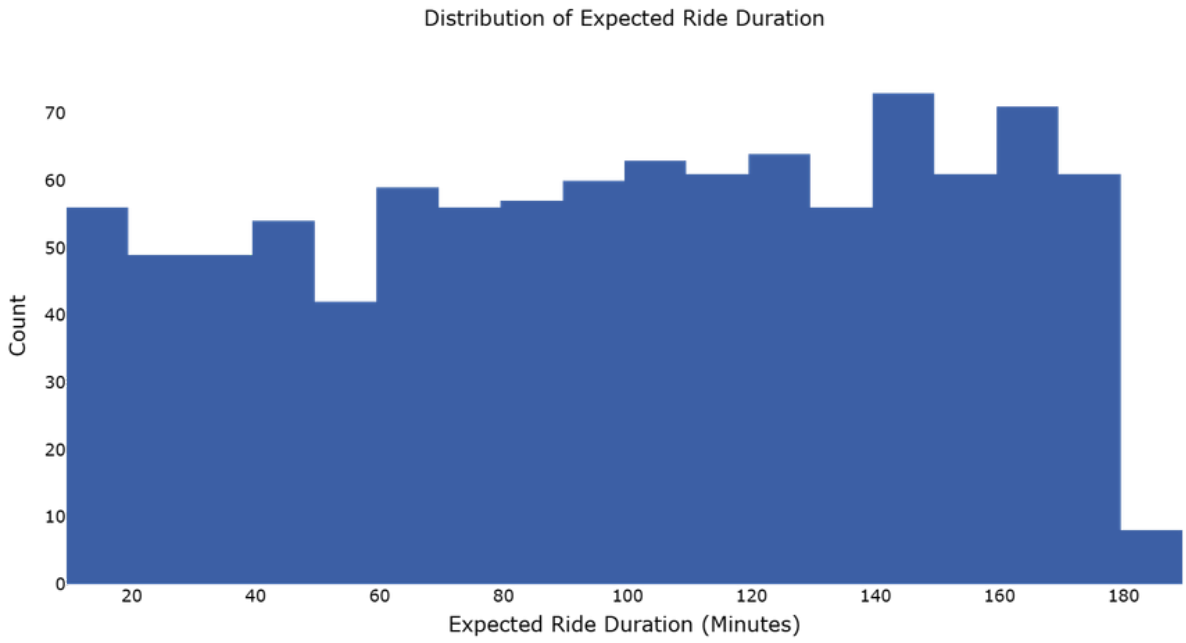
Key observations

- Historical ride costs span approximately $25 to $836, with the highest concentration of observations occurring roughly between $150 and $500.
- The distribution is reasonably broad, with observations extending across both lower-cost and higher-cost rides rather than being concentrated around a single narrow price range.

Business Insights

- The wide cost range indicates that the pricing model needs to account for multiple ride characteristics rather than relying on a single fixed fare level.
- The concentration of rides in the mid-cost range provides a substantial base of observations for learning typical pricing patterns, while higher-cost rides require the model to handle greater variation.

Expected ride durations are distributed across a broad operating range, with observations spread relatively consistently across most durations.



Key observations

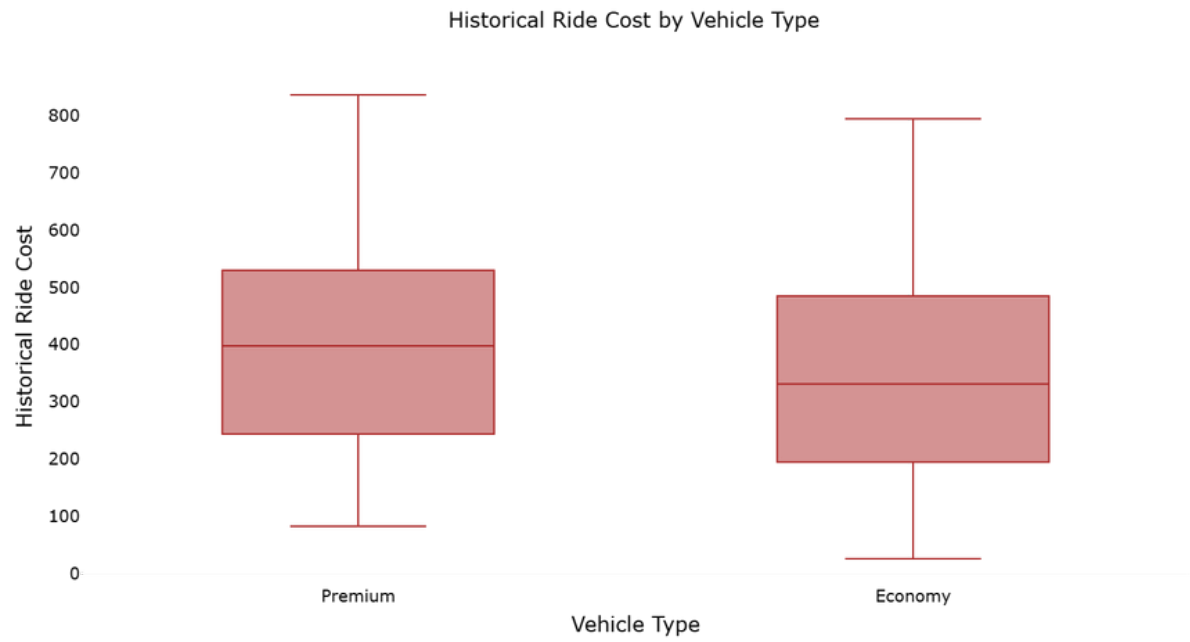
- Expected ride duration ranges from approximately 10 to 180 minutes, with observations distributed across most of this range.
- The distribution does not show a strong concentration around one duration; several intervals between approximately 60 and 175 minutes contain relatively high observation counts.

Business Insights

- The broad duration range provides the model with observations covering both shorter and longer rides, supporting the analysis of duration-related pricing patterns.
- Since ride duration varies substantially across trips, incorporating Expected_Ride_Duration into the pricing model is important for capturing differences in historical ride costs.

Premium rides generally have higher historical costs than Economy rides, while both vehicle types show substantial cost variation.

~$70 higher
Premium Median Cost



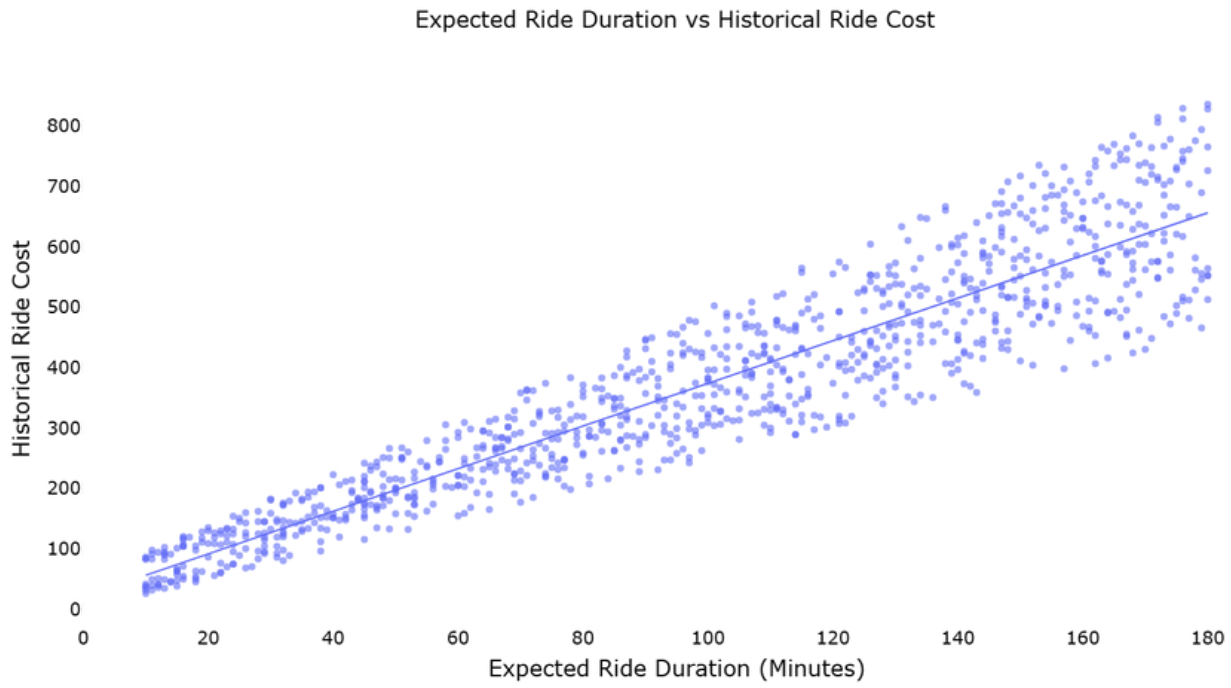
Key observations

- The median historical cost for Premium rides is visibly higher than the median for Economy rides, with the Premium median around $400 compared with roughly $330 for Economy.
- Both vehicle types have wide cost ranges, extending from relatively low-cost rides to values approaching or exceeding $800, indicating that vehicle type alone does not explain the full variation in historical ride costs.

Business Insights

- Vehicle type should be incorporated into pricing models because the observed cost distributions differ between Premium and Economy rides.
- The substantial variation within each vehicle category indicates that pricing decisions should combine vehicle type with other ride characteristics rather than applying a fixed vehicle-specific price.

Historical ride cost shows a strong positive relationship with expected ride duration



Key observations

- Historical ride cost increases consistently as expected ride duration increases, with the plotted relationship showing a strong positive trend across the full duration range.
- The spread of ride costs becomes wider at longer durations, particularly beyond approximately 120 minutes, indicating greater variation in historical costs for longer rides.

Business Insights

- Expected ride duration should be an important input in ride-cost prediction because longer rides are consistently associated with higher historical costs.
- The increasing variation at longer durations indicates that additional ride characteristics are needed to explain differences in cost beyond duration alone.

Driver availability generally increases as the number of riders increases, while supply variation becomes wider at higher rider volumes



Key observations

- The scatter plot shows a clear positive relationship between the number of riders and the number of drivers, indicating that higher rider volumes are generally accompanied by greater driver availability.
- At higher rider counts, the range of observed driver counts becomes substantially wider, showing greater variation in available supply under higher demand conditions.

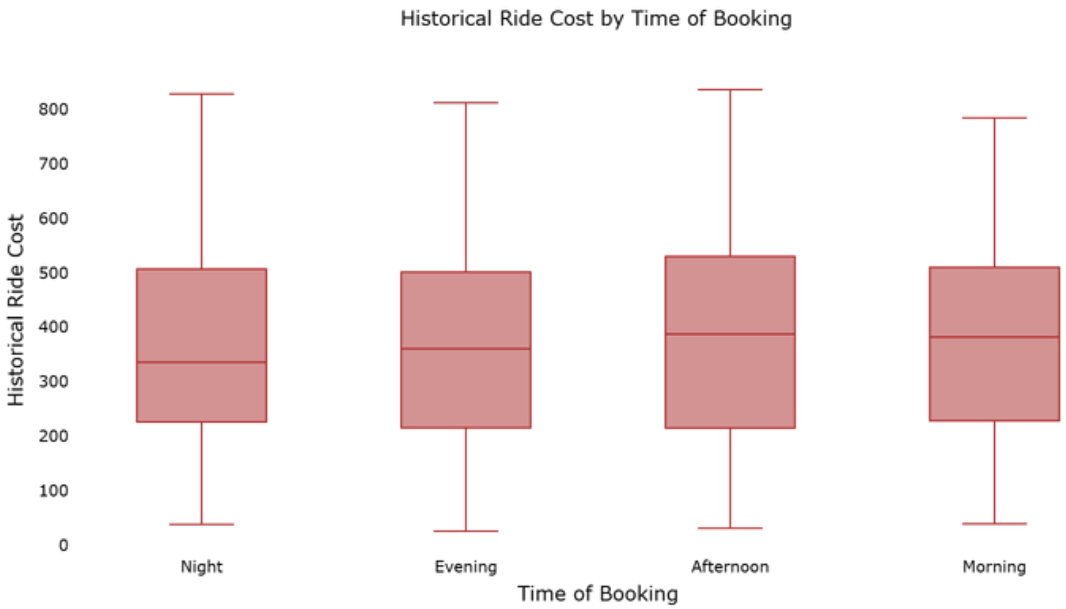
Business Insights

- Pricing analysis should consider both rider demand and driver supply rather than treating either variable independently.
- The variation in driver availability at higher rider volumes supports the use of a combined demand-supply feature such as Rider_Driver_Ratio to represent the balance between riders and drivers.

Historical ride costs show relatively similar distributions across booking periods, with only moderate differences in median cost.

Afternoon

Highest Median Cost



Key observations

- The median historical cost is highest for Afternoon bookings, followed closely by Morning, while Night has the lowest median among the four periods.
- The distributions overlap substantially across all four booking periods, and each period shows a wide range of historical ride costs.

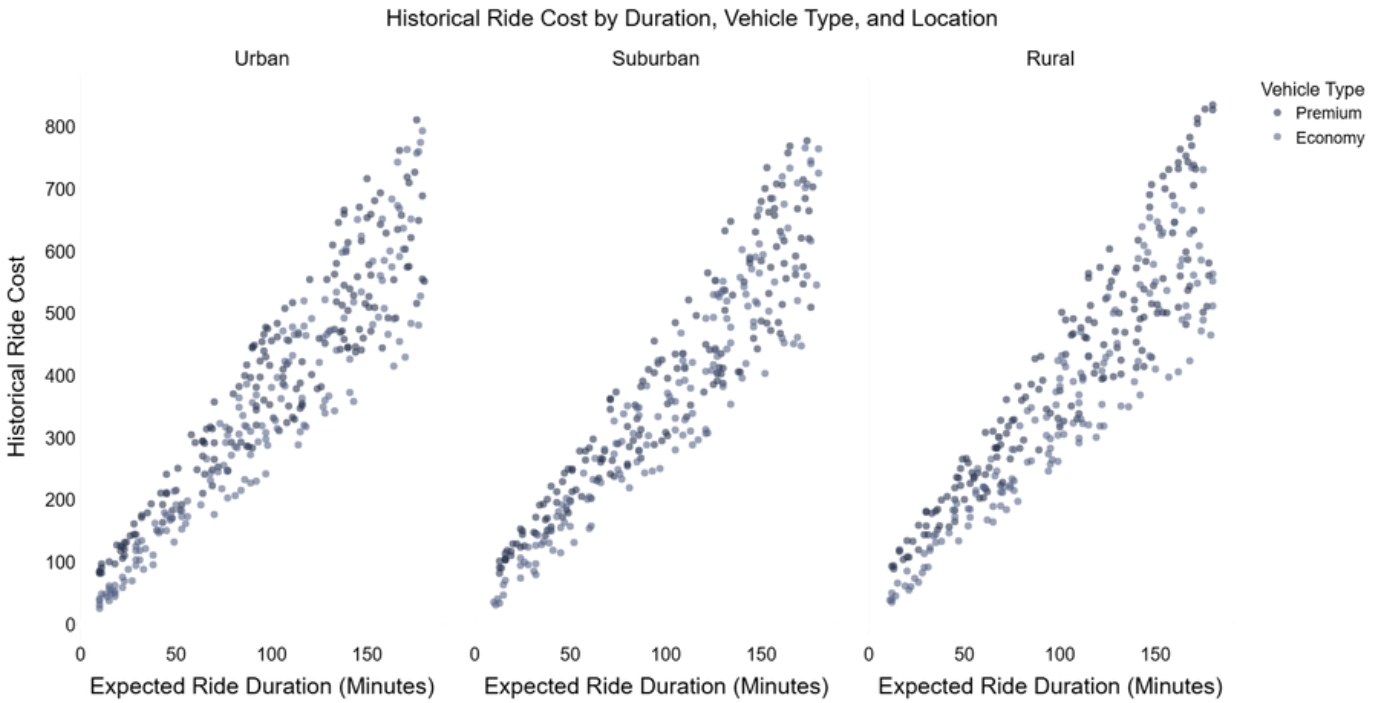
Business Insights

- Time of booking can be retained as a pricing-related feature, but the substantial overlap suggests that booking period alone does not determine historical ride cost.
- Pricing analysis should combine booking time with stronger ride-level predictors such as expected duration and vehicle type rather than applying substantially different pricing solely by time period.

The positive duration-cost relationship remains consistent across vehicle types and locations, with Premium rides generally positioned above Economy rides

Premium > Economy

Higher Cost at Similar Durations



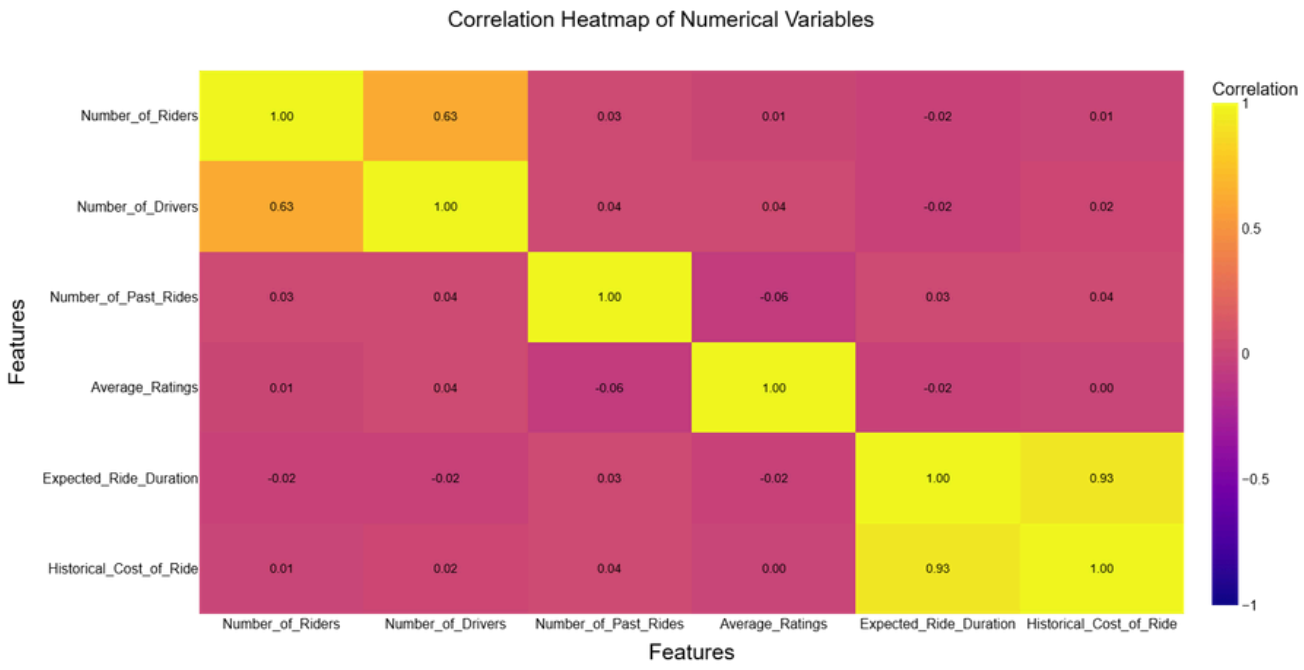
Key observations

- Historical ride cost increases with expected ride duration across Urban, Suburban, and Rural locations, showing a consistent positive relationship in all three panels.
- Within each location, Premium rides generally show higher historical costs than Economy rides at comparable ride durations, although the two groups overlap.

Business Insights

- Ride duration, vehicle type, and location should be considered together when analyzing historical ride costs rather than evaluating each factor independently.
- The consistent pattern across locations suggests that the relationship between duration and historical cost is not limited to a single geographic category.

Expected ride duration has the strongest numerical relationship with historical ride cost, while rider and driver counts show a moderate relationship with each other.



Key Observations

- Expected_Ride_Duration has a very strong positive correlation with Historical_Cost_of_Ride of approximately 0.93, making it the strongest numerical relationship with the target.
- Number_of_Riders and Number_of_Drivers have a moderate positive correlation of approximately 0.63, while the remaining numerical variables show correlations close to zero with historical ride cost.

Business Insights

- Expected ride duration should receive substantial attention in ride-cost modeling because of its strong observed relationship with historical ride cost.
- The moderate rider-driver relationship supports examining demand and supply together, which is consistent with the engineered Rider_Driver_Ratio feature used during model preparation.

Data Preparation & Feature Engineering



3.1 DATA PREPARATION

Following the exploratory analysis, the dataset was prepared for machine learning.

The preparation process included:

- Selecting the target variable.
- Separating predictor variables from the target.
- Identifying numerical and categorical variables.
- Creating the demand-supply feature.
- Splitting the dataset into training and testing portions.
- Preparing numerical variables for scaling.
- Preparing categorical variables for one-hot encoding.
- Saving the processed datasets and preprocessing artifact for subsequent notebooks.

The target variable was:

Historical_Cost_of_Ride

The final dataset was divided using an 80/20 train-test split.

- Training observations: 800
- Testing observations: 200

A random_state of 42 was used to ensure reproducibility.



3.2 DEMAND-SUPPLY FEATURE ENGINEERING

The EDA demonstrated a relationship between the number of riders and available drivers.

To represent demand relative to available supply, the following feature was created:

Rider_Driver_Ratio

The feature was calculated as:

Rider_Driver_Ratio = Number_of_Riders / Number_of_Drivers

This feature provides a direct representation of the number of riders relative to available drivers.

The feature was retained as an additional numerical predictor for modeling



3.3 CATEGORICAL ENCODING

The dataset contains four categorical variables:

- *Location_Category*
- *Customer_Loyalty_Status*
- *Time_of_Booking*
- *Vehicle_Type*

These variables were transformed using One-Hot Encoding.

The encoder was configured with handle_unknown='ignore' to ensure that previously unseen categories would not cause errors during prediction.

3.4 NUMERICAL FEATURE SCALING

The numerical variables were standardized using StandardScaler.

- *Number_of_Riders*
- *Number_of_Drivers*
- *Number_of_Past_Rides*
- *Average_Ratings*
- *Expected_Ride_Duration*
- *Rider_Driver_Ratio*

The scaler was fitted using the training data and then applied to both training and testing data.



3.5 TRAIN-TEST SPLIT

The prepared dataset was divided into:

- 80% training data: 800 observations
- 20% testing data: 200 observations

The training data was used for model fitting and hyperparameter optimization, while the testing data was reserved for final evaluation.

After preprocessing and one-hot encoding, the dataset contained 18 processed features.

The processed datasets were saved as:

- X_train.csv
- X_test.csv
- y_train.csv
- y_test.csv

The fitted preprocessing pipeline was also saved for reuse during model evaluation and explainability.



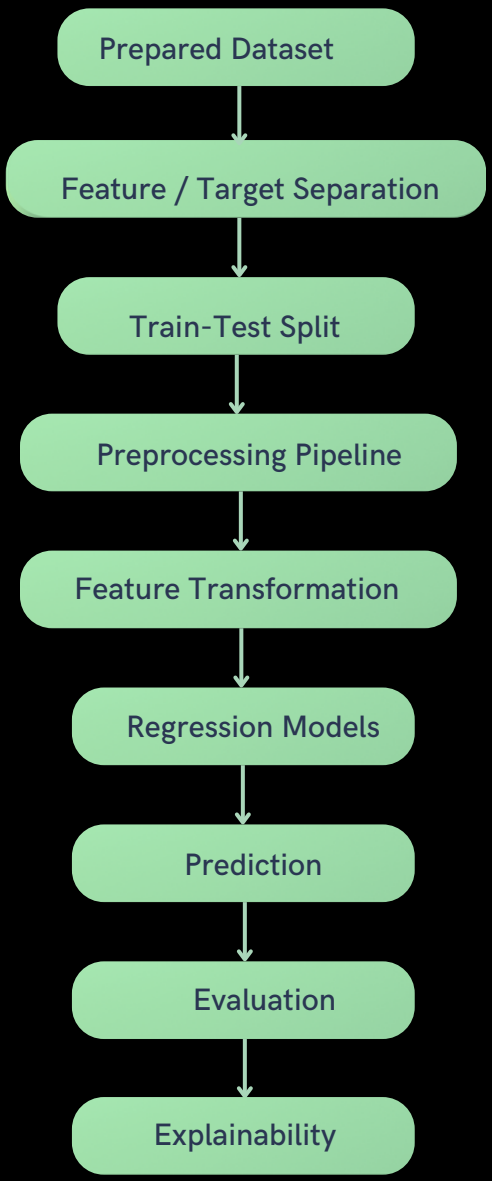
Model Development & Experimentation



4.1 MODELING APPROACH

The problem was formulated as a supervised regression problem, with ***Historical_Cost_of_Ride*** as the continuous target variable.

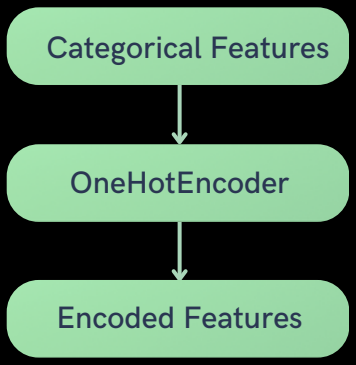
The modeling workflow consisted of:



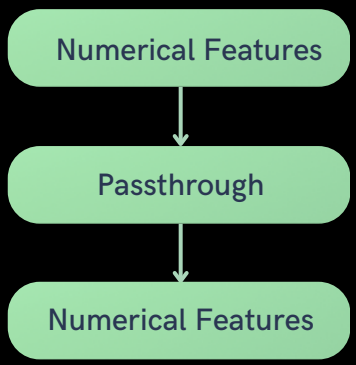
4.2 PREPROCESSING PIPELINE

A unified preprocessing pipeline was used to ensure that the same transformations were applied consistently during training and prediction.

Categorical Pipeline



Numerical Pipeline



Both were combined through a *ColumnTransformer*.

The fitted preprocessor was saved as a reusable artifact to maintain consistency across subsequent evaluation and explainability notebooks.

4.3 REGRESSION MODELING

Multiple regression approaches were developed and evaluated.

Initial Models

- Linear Regression
- Ridge Regression
- Lasso Regression
- Random Forest
- Extra Trees
- Gradient Boosting
- MLP Regressor
- XGBoost
- LightGBM

Tuned Models

Hyperparameter optimization was performed for:

- Random Forest
- Extra Trees
- Gradient Boosting
- XGBoost
- LightGBM



.. continued.

A lightweight RandomizedSearchCV configuration was used with:

- 8 parameter combinations
- 3-fold cross-validation
- 24 fits per model
- n_jobs=1
- random_state=42

This configuration was selected to provide meaningful hyperparameter exploration while keeping computational requirements practical.

Ensemble Models

Two ensemble approaches were also developed:

- Voting Regression
- Stacking Regression

Two ensemble approaches were also developed:

- Ridge Regression
- Gradient Boosting
- XGBoost
- LightGBM

with a final Ridge Regression estimator.

Baseline Model

A mean-based DummyRegressor was used as the baseline.

- MAE: 160.9642
- RMSE: 191.1528
- R^2: -0.0022

This baseline provided a reference point for determining whether the developed models learned useful predictive relationships.

Model Evaluation & Comparison

5.1 EVALUATION METRICS

The developed models were evaluated using three primary regression metrics.

Mean Absolute Error (MAE)

- MAE measures the average absolute difference between actual and predicted historical ride costs.
- A lower MAE indicates that predictions are closer to actual values on average.

Root Mean Squared Error (RSME)

- RMSE gives greater weight to larger prediction errors.
- This makes RMSE useful for identifying models that produce relatively large prediction errors.

R^2 Score

- R^2 measures the proportion of variation in historical ride costs explained by the model.
- A higher R^2 indicates stronger explanatory performance on the evaluated dataset.



5.2 ERROR ANALYSIS

Beyond the overall evaluation metrics, prediction errors were examined through:

- Residual analysis
- Actual vs predicted analysis
- Baseline comparison
- Maximum absolute error analysis
- Model-level error comparison

The evaluation was performed on the held-out test dataset containing 200 observations.



5.2.1 RESIDUAL ANALYSIS

Residuals were calculated as:

Actual - Predicted

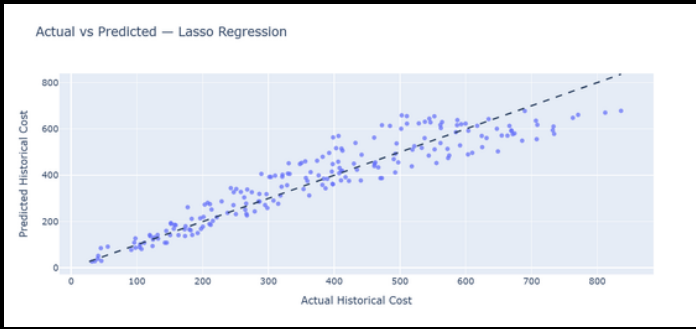
This indicates that model performance varies depending on product category.

The negative mean errors observed across the models indicate a slight tendency toward over-prediction.

However, individual residuals were considerably larger for some observations.

The maximum absolute errors varied across models, demonstrating that strong aggregate performance does not eliminate large errors for individual rides.

5.2.2 ACTUAL VS PREDICTED ANALYSIS









The actual-versus-predicted analysis showed that predictions generally align with observed historical ride costs.

Most observations were positioned relatively close to the reference line, indicating that the models captured a substantial proportion of the variation in ride costs.

However, deviations from the reference line remained visible for individual observations, demonstrating that some ride costs were more difficult to estimate accurately.

5.3 LARGEST PREDICTION ERRORS

The largest individual error investigated in detail occurred for the tuned Gradient Boosting model.

Metric	Value
Actual Historical Cost	398.32
Predicted Historical Cost	598.44
Absolute Error	200.12

The model therefore over-predicted the historical ride cost by approximately 200.12 for this observation.

The corresponding ride characteristics were:

- Number of Riders: 59
- Number of Drivers: 40
- Location Category: Rural
- Customer Loyalty Status: Gold
- Number of Past Rides: 64
- Average Ratings: 4.72
- Time of Booking: Night
- Vehicle Type: Economy
- Expected Ride Duration: 157 minutes
- Rider_Driver_Ratio: 1.475
- Actual Cost: 398.32
- Predicted Cost: 598.44



5.4 INVESTIGATION OF THE LARGEST ERROR

The selected observation represents a Rural, Economy-vehicle ride booked at Night.

- 59 riders
- 40 drivers
- A rider-driver ratio of 1.475
- 64 past rides
- An average rating of 4.72
- An expected ride duration of 157 minutes

Despite the long expected ride duration, the historical ride cost was only approximately 398.32.

The tuned Gradient Boosting model predicted approximately 598.44.

Explainability Conclusion

The model's prediction was strongly influenced by expected ride duration, resulting in a substantial over-prediction for this particular ride.

- The individual SHAP explanation showed that Expected_Ride_Duration contributed approximately +226.52 to the model prediction.
- Other features provided smaller positive and negative adjustments, but their combined contributions did not sufficiently offset the strong contribution associated with expected ride duration.
- This demonstrates that the model learned a strong relationship between expected ride duration and historical ride cost, but that relationship does not perfectly describe every individual observation.



5.5 UNTUNED VS TUNED MODEL COMPARISON

Hyperparameter tuning produced mixed results across the five non-linear model families.

Model	Untuned RMSE	Value
Random Forest	74.6436	78.9125
Extra Trees	72.0363	83.8946
Gradient Boosting	72.1217	69.9215
XGBoost	71.9309	73.5117
LightGBM	79.1346	72.1934

Interpretation

Gradient Boosting and LightGBM achieved lower test-set RMSE after tuning.

Random Forest, Extra Trees, and XGBoost produced higher test-set RMSE after tuning.

This demonstrates that hyperparameter tuning does not automatically improve performance on unseen data. The effectiveness of tuning depends on the model, search space, and selected parameter configuration.



5.6 ENSEMBLE MODEL COMPARISON

The ensemble models demonstrated different performance characteristics.

Voting Regression

- MAE: 52.9107
- RMSE: 69.6303
- R^2: 0.8670

Voting Regression

- MAE: 52.0649
- RMSE: 67.7824
- R^2: 0.8740

Stacking Regression achieved a lower MAE than the individual models included in the ensemble, while its RMSE remained close to the strongest individual models.

Voting Regression produced a higher RMSE than the strongest individual models.

The results demonstrate that combining multiple models does not automatically produce lower error across every evaluation metric.

5.7 OVERALL MODEL COMPARISON

The final evaluation included 16 trained models.

Model	MAE	RMSE	R^2
Lasso Regression	**52.6056**	**67.5878**	**0.8747**
Linear Regression	**52.6138**	**67.5885**	**0.8747**
Ridge Regression	**52.5943**	**67.5893**	**0.8747**
Stacking Regression	**52.0649**	**67.7824**	**0.8740**
Voting Regression	**52.9107**	**69.6303**	**0.8670**
MLP Regressor	**55.5282**	**69.8909**	**0.8660**
Gradient Boosting Tuned	**53.3413**	**69.9215**	**0.8659**
XGBoost	**54.2290**	**71.9309**	**0.8581**
Extra Trees	**53.0204**	**72.0363**	**0.8577**
Gradient Boosting	**53.6769**	**72.1217**	**0.8573**
LightGBM Tuned	**54.7664**	**72.1934**	**0.8571**



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Model	MAE	RMSE	R^2
XGBoost Tuned	**54.7342**	**73.5117**	**0.8518**
Random Forest	**55.6138**	**74.6436**	**0.8472**
Random Forest Tuned	**63.3565**	**78.9125**	**0.8292**
LightGBM	**57.9706**	**79.1346**	**0.8282**
Extra Trees Tuned	**66.9269**	**83.8946**	**0.8070**

The strongest-performing models were closely grouped across the three evaluation metrics.

- ***Lowest MAE***: Stacking Regression - 52.0649
- ***Lowest RMSE***: Lasso Regression - 67.5878
- ***Highest R^2***: Lasso Regression - 0.8747

Therefore, no single model records the lowest error across all three evaluation metrics.



Explainability & Final Model

6.1 WHY EXPLAINABILITY WAS INCLUDED

Accurate predictions alone are not always sufficient for business applications.

Retail decision-makers also need to understand:

- Which variables influence predictions?
- Which ride characteristics matter most?
- What drives model behavior?
- Where does the model struggle?
- Why can individual predictions differ substantially from actual historical costs?

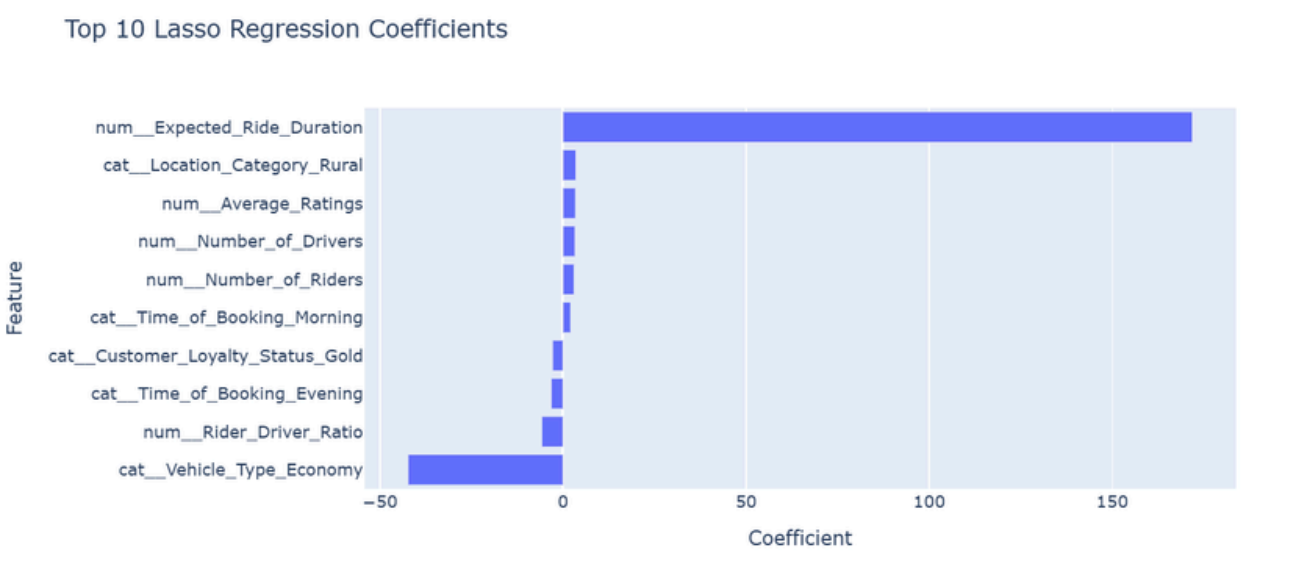
Explainability analysis was therefore performed to understand the model's prediction behavior.

The analysis included:

- Lasso Regression coefficients
- SHAP analysis
- SHAP directional analysis
- Permutation feature importance
- Individual prediction analysis
- Stacking component analysis



6.2 FEATURE IMPORTANCE



The highest-ranked model features were:

Feature	Value
Expected_Ride_Duration	34.22
Vehicle_Type_Economy	**22.52**
Rider_Driver_Ratio	**11.25**
Location_Category_Rural	**3.47**
Average_Ratings	**2.41**
Number_of_Drivers	**2.40**

.. continued.

Feature	Value
Time_of_Booking_Evening	34.22
Number_of_Riders	22.52
Customer_Loyalty_Status_Gold	11.25
Time_of_Booking_Morning	3.47

Expected_Ride_Duration has the largest absolute coefficient by a substantial margin.

Lasso also assigns zero or effectively zero coefficients to several encoded features, including:

- Location_Category_Suburban
- Customer_Loyalty_Status_Regular
- Vehicle_Type_Premium

This demonstrates the feature-selection effect of Lasso regularization.

The coefficient values describe the model's learned relationships within the processed feature space and should not be interpreted as causal effects.

6.3 SHAP FEATURE IMPORTANCE

SHAP analysis was performed on the tuned Gradient Boosting model.

The highest mean absolute SHAP values were:

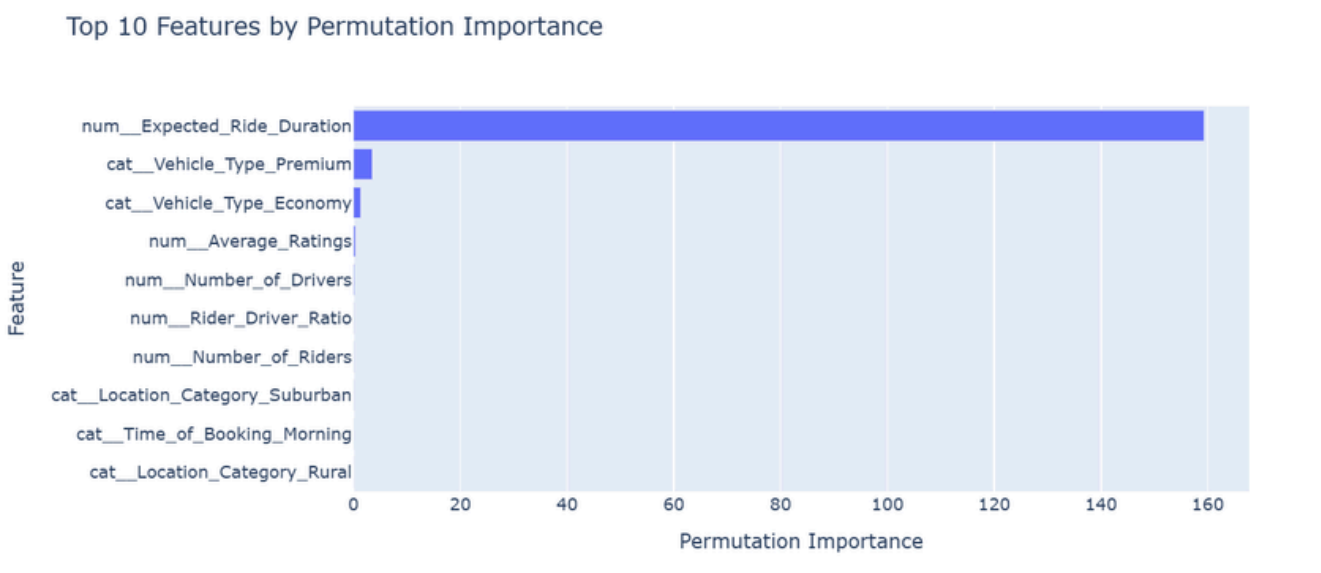
Rank	Feature	Value
1	Expected_Ride_Duration	150.3693
2	Vehicle_Type_Premium	10.4511
3	Vehicle_Type_Economy	5.9326
4	Rider_Driver_Ratio	3.8213
5	Number_of_Drivers	3.6967
6	Average_Ratings	3.5726
7	Number_of_Riders	1.1874
8	Number_of_Past_Rides	0.8486
9	Location_Category_Rural	0.4876
10	Customer_Loyalty_Status_Gold	0.3380

Expected_Ride_Duration is clearly the dominant feature, with a mean absolute SHAP value of approximately 150.37.

Vehicle type provides the next notable contribution, followed by demand-supply and other operational characteristics.



6.4 PERMUTATION FEATURE IMPORTANCE



Permutation feature importance was calculated for the tuned Gradient Boosting model using MAE as the scoring measure.

The highest-ranked features were:

Rank	Feature	Importance
1	Expected_Ride_Duration	159.3334
2	Vehicle_Type_Premium	3.5364
3	Vehicle_Type_Economy	1.3668
4	Average_Ratings	0.3688
5	Number_of_Drivers	0.1839

.. continued.

Rank	Feature	Importance
6	Rider_Driver_Ratio	0.0792
7	Number_of_Riders	0.0576
8	Location_Category_Suburban	0.0513
9	Time_of_Booking_Morning	0.0365
10	Location_Category_Rural	0.0077

The permutation analysis provides a result broadly consistent with SHAP.

Expected Ride Duration remains the dominant feature, while vehicle type provides the next notable contribution.



6.5 INDIVIDUAL PREDICTION EXPLAINABILITY

The largest error investigated earlier was also analyzed using SHAP.

The selected observation had:

- Actual cost: 398.32
- Predicted cost: 598.44
- Absolute error: 200.12

The largest positive SHAP contributions were:

Feature	SHAP Contribution
Expected_Ride_Duration	+226.52
Average_Ratings	**+7.12**
Rider_Driver_Ratio	+5.87
Number_of_Drivers	**+2.19**
Number_of_Past_Rides	**+1.12**

.. continued.

The strongest negative contributions included:

Feature	SHAP Contribution
Vehicle_Type_Premium	-9.74
Vehicle_Type_Economy	-5.30
Customer_Loyalty_Status_Gold	-0.74
Number_of_Riders	-0.19
Customer_Loyalty_Status_Silver	-0.08

The analysis shows that expected ride duration contributed substantially more to the prediction than the other features for this particular observation.

This explains why the model produced a high predicted cost despite the actual historical cost being considerably lower.

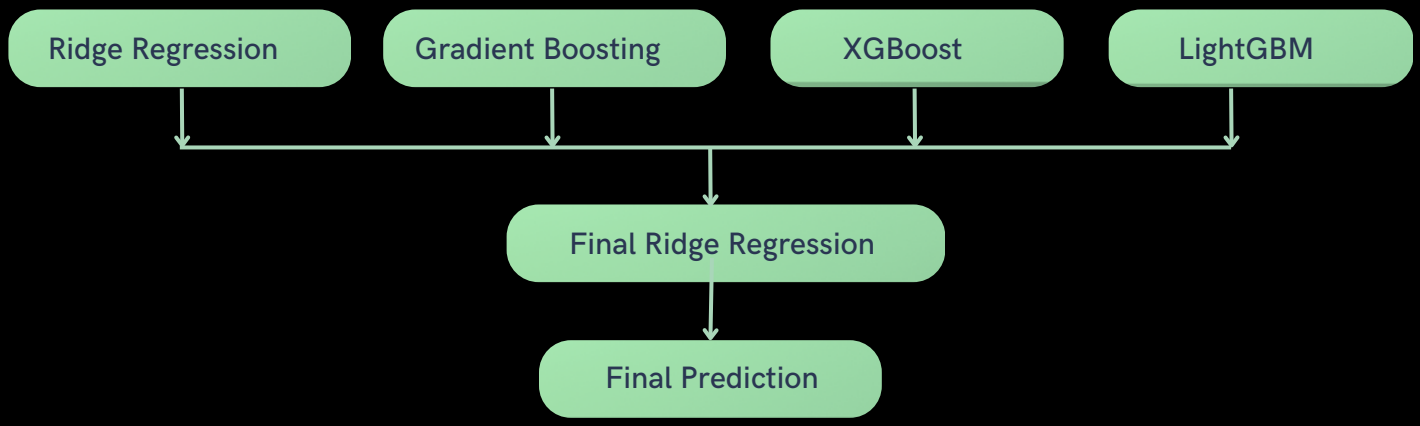


6.6 STACKING REGRESSION COMPONENTS

The Stacking Regression model combines four base estimators:

- Ridge Regression
- Gradient Boosting
- XGBoost
- LightGBM

The predictions from these base models are passed to a final Ridge Regression estimator.



This structure combines linear and non-linear modeling approaches within a single ensemble.

The component analysis provides a model-level explanation of how the stacking model generates predictions, while the Lasso, SHAP, and permutation analyses provide feature-level explanations.

6.7 EXPLAINABILITY SUMMARY

The explainability analysis demonstrates several important characteristics of the developed models.

Strength

Expected Ride Duration is consistently identified as the strongest predictive signal across:

- Lasso coefficients
- SHAP analysis
- Permutation feature importance

Vehicle type, demand-supply characteristics, driver availability, and average ratings provide additional predictive information.

Model Behavior

The consistency between SHAP and permutation importance strengthens the interpretation that expected ride duration plays a dominant role in the current dataset.

Model Limitation

The individual prediction analysis demonstrates that strong overall model performance does not eliminate substantial errors for specific rides.

In the investigated high-error observation, the model substantially over-predicted historical ride cost because the learned relationship between expected ride duration and cost was stronger than the actual relationship for that particular observation.



Business Insights & Recommendations

7.1 RIDE DURATION & FARE ESTIMATION

Insight

Expected Ride Duration is the strongest and most consistent predictive signal identified throughout the analysis.

It has a correlation of approximately 0.93 with historical ride cost and is also the most influential feature in the Lasso, SHAP, and permutation analyses.

Recommendation

Ride-duration information should remain a central component of ride-cost estimation.

However, duration should not be considered in isolation because substantial variation remains among rides with similar durations.



7.2 DEMAND-SUPPLY CONDITIONS

Insight

Number of Riders and Number of Drivers have a moderate positive correlation of approximately 0.63.

The Rider_Driver_Ratio feature provides an additional representation of rider demand relative to driver availability.

Recommendation

Demand and supply should be monitored jointly when analyzing ride-pricing conditions.

A demand-supply ratio can provide an additional operational signal alongside absolute rider and driver counts.

7.3 VEHICLE-BASED PRICING

Insight

Premium rides have a higher median historical cost than Economy rides.

The explainability analysis also identifies vehicle type as an important feature after expected ride duration.

Recommendation

Vehicle category should be incorporated into ride-cost estimation rather than relying solely on duration.

Pricing analysis should also account for the variation within each vehicle category.



7.4 BOOKING-TIME PATTERNS

Insight

Afternoon bookings have the highest median historical cost, followed by Morning bookings, while Night bookings have the lowest median.

However, the distributions across booking periods overlap considerably.

Recommendation

Booking period can be incorporated as a supporting pricing feature.

It should be combined with stronger signals such as expected duration and vehicle type rather than being treated as an independent pricing rule.

7.5 CUSTOMER & RIDE CHARACTERISTICS

Insight

Average ratings, customer loyalty, and number of past rides contribute additional information to model predictions, but their influence is considerably smaller than expected ride duration and vehicle type.

Recommendation

Customer and historical ride characteristics can be retained as supporting variables within predictive models.

These variables should complement operational pricing signals rather than replace them.



7.6 PREDICTION ERROR MANAGEMENT

Insight

The models achieve strong aggregate performance, but individual prediction errors can still be substantial.

The largest investigated error involved an over-prediction of approximately 200.12.

Recommendation

Ride-cost prediction systems should monitor individual prediction errors in addition to aggregate metrics.

Large deviations can be investigated to identify ride conditions that are not sufficiently represented by the available features.

7.7 DATA-DRIVEN PROCUREMENT

Insight

The developed models demonstrate that historical ride costs can be estimated using a combination of ride duration, vehicle type, demand-supply information, and other operational characteristics.

However, the current dataset does not contain a dedicated optimal-price or surge-pricing target.

Recommendation

The current model can serve as a foundation for historical ride-cost analysis, but a production dynamic-pricing system would require additional real-time variables and an explicitly defined pricing objective.



Conclusion

8.1 PROJECT SUMMARY

The Ride Pricing Analytics project demonstrates how machine learning can be applied to ride-sharing data to predict historical ride costs and understand the factors influencing model predictions.

The project followed a complete machine learning workflow covering:

- Data understanding
- Data validation
- Exploratory Data Analysis
- Outlier analysis
- Feature engineering
- Data preprocessing
- Regression modeling
- Hyperparameter tuning
- Ensemble learning
- Model evaluation
- Model comparison
- Residual analysis
- Model explainability
- Business interpretation

A total of 16 models were evaluated against a mean-based baseline.

8.2 KEY FINDINGS

The analysis identified several important findings:

- Historical ride costs range broadly from approximately 25.99 to 836.12.
- Expected Ride Duration has a very strong positive relationship with historical ride cost, with a correlation of approximately 0.93.
- Premium rides generally have higher historical costs than Economy rides.
- Number of Riders and Number of Drivers show a moderate positive correlation of approximately 0.63.
- Booking periods show substantial overlap despite differences in median ride costs.
- Rider_Driver_Ratio provides an additional representation of demand relative to driver availability.
- Expected Ride Duration is consistently the dominant predictive feature across Lasso, SHAP, and permutation importance analysis.
- Vehicle type provides additional predictive information beyond ride duration.
- All trained models substantially outperform the mean-based baseline.
- Lasso Regression records the lowest RMSE and highest R^2.
- Stacking Regression records the lowest MAE.
- Hyperparameter tuning produces mixed results across the evaluated model families.
- Individual prediction analysis demonstrates that substantial errors can remain despite strong overall test-set performance.

8.3 OVERALL MODEL ASSESSMENT

The developed models provide a useful framework for estimating historical ride costs from ride-level, customer, vehicle, booking, and demand-supply information.

The strongest results are concentrated among Lasso Regression, Linear Regression, Ridge Regression, and Stacking Regression.

Rank	MAE	RMSE	R^2
Lasso Regression	**52.6056**	**67.5878**	0.8747
Linear Regression	**52.6138**	**67.5885**	**0.8747**
Ridge Regression	**52.5943**	**67.5893**	**0.8747**
Stacking Regression	**52.0649**	**67.7824**	**0.8740**

Lasso Regression records the lowest RMSE and highest R^2, while Stacking Regression records the lowest MAE. Linear Regression and Ridge Regression produce very similar results across all three metrics.

Therefore, there is no single model that produces the lowest value across all three evaluation metrics. The results instead show that different models provide very similar predictive performance, with their relative position depending on the evaluation metric being considered.

8.4 LIMITATIONS

1. Dataset Size

The dataset contains only 1,000 observations, which limits the amount of information available for learning more complex pricing patterns.

2. Dataset Representation

The target represents historical observed ride cost rather than an explicitly defined optimal dynamic price.

Therefore, the model estimates historical pricing behavior rather than determining the objectively optimal fare.

3. Limited Real-Time Information

The dataset does not include several variables that would normally influence real-time ride pricing, including:

- Real-time surge pricing
- Traffic conditions
- Weather conditions
- Geographic demand concentration
- Local events
- Detailed temporal pricing patterns
- Real-time driver availability



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4. Dominance of Ride Duration

Expected Ride Duration is substantially more influential than the other available variables.

This suggests that the current dataset may not contain enough additional information to fully represent real-world dynamic pricing behavior.

5. Individual Prediction Errors

Strong aggregate performance does not eliminate substantial errors for individual rides.

The investigated high-error observation demonstrated that the model can significantly overestimate historical ride cost under certain feature combinations.



8.5 FUTURE SCOPE

1. Larger and More Diverse Data

Introduce larger ride-sharing datasets containing more observations across locations, time periods, vehicles, and demand conditions.

2. Real-Time Demand and Supply

Incorporate real-time rider demand and driver availability to better represent changing market conditions.

3. Geographic Features

Add more detailed geographic information such as:

- Pickup location
- Drop-off location
- Distance
- Geographic demand concentration
- Route characteristics

4. External Conditions

Incorporate:

- Traffic conditions
- Weather
- Holidays
- Events
- Local demand indicators



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5. Dynamic Pricing Target

Develop a dedicated target representing an optimal fare, surge multiplier, or pricing decision rather than relying only on historical observed cost.

6. Time-Aware Modeling

With sufficiently detailed chronological data, evaluate time-aware validation and forecasting approaches to better represent changing pricing behavior.

6. Production Deployment

The model could eventually form part of a broader pricing decision-support system providing:

- Historical cost estimation
- Demand-supply monitoring
- Ride-cost anomaly detection
- Prediction-confidence monitoring
- Pricing analytics
- Real-time dynamic pricing support

The current project therefore provides a foundation for understanding historical ride-cost behavior while clearly identifying the additional data and modeling capabilities required for a production-oriented dynamic pricing system.



PROJECT SUMMARY

This project provides a data-driven view of ride pricing by combining exploratory analysis, feature engineering, predictive modeling, model evaluation, and explainability. The findings highlight how expected ride duration, vehicle type, demand-supply conditions, driver availability, ratings, and booking characteristics contribute to historical ride-cost prediction. Model evaluation demonstrates that different regression approaches perform differently across MAE, RMSE, and R^2, while explainability analysis further identifies expected ride duration as the most consistent predictive signal. The project establishes a foundation for extending mobility pricing analytics through richer demand-supply features, advanced pricing models, improved prediction accuracy, and real-time ride pricing applications.

This project was developed using Python, Pandas, NumPy, Matplotlib, Seaborn, Plotly, Scikit-learn, XGBoost, LightGBM, MLflow and SHAP.

Author:

Tejas Jadhav

GitHub: **@tejas-jadhav**

LinkedIn: **Tejas Jadhav**